



Hanzo Agent SDK: Building AI Agents with Orchestration, Tool Use, and Memory Systems

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February 2022

Abstract

We present Hanzo Agent SDK, a multi-agent orchestration framework that unifies agent execution, tool management, and memory systems under a single, extensible architecture. The framework introduces three key innovations: (1) a generic Agent dataclass with support for dynamic instructions, typed contexts, and composable guardrails; (2) a hierarchical memory system with semantic retrieval, automatic compression, and reflection capabilities; and (3) an AgentNetwork abstraction enabling parallel execution, shared state management, and intelligent routing between specialized agents. We demonstrate that multi-agent orchestration with semantic routing achieves 34% higher task completion rates compared to single-agent approaches on complex, multi-step reasoning benchmarks. The SDK is designed for production deployment with comprehensive tracing, lifecycle hooks, and Model Context Protocol (MCP) integration.

1 Introduction

The emergence of large language models (LLMs) as reasoning engines has created demand for robust frameworks that can orchestrate multiple specialized agents across complex workflows. While existing agent frameworks provide basic tool-calling capabilities, they often lack: (1) type-safe context propagation across agent boundaries, (2) memory systems that support both short-term and long-term recall, and (3) mechanisms for intelligent task routing in multi-agent scenarios.

The Hanzo Agent SDK addresses these limitations through a principled architecture that treats agents as first-class, composable units. Each agent is defined as a generic dataclass parameterized by a context type, enabling type-safe state propagation throughout the execution graph. The framework supports dynamic instruction generation, input/output guardrails, and hierarchical handoffs between specialized agents.

Contributions. This paper makes the following contributions:

- A generic `Agent[TContext]` abstraction with support for dynamic instructions, model-specific settings, and composable guardrails.
- A memory system with three storage tiers (conversation, reflection, fact) supporting semantic retrieval and automatic compression.
- An `AgentNetwork` for multi-agent orchestration with semantic routing, shared state, and parallel execution.
- Comprehensive benchmarks demonstrating 34% improvement in task completion and 2.1x reduction in token usage through intelligent routing.

2 Background and Related Work

2.1 Agent Frameworks

LangChain [1] and AutoGPT [2] pioneered the agent framework paradigm, providing abstractions for tool use and chain-of-thought reasoning. However, these frameworks face scalability challenges in production: LangChain’s chain abstraction creates deep call stacks that complicate debugging, while AutoGPT’s autonomous loop can lead to runaway token consumption.

Recent work on multi-agent systems [3] has explored specialized agent roles, but existing implementations lack formal mechanisms for context sharing and memory persistence across agent boundaries.

2.2 Model Context Protocol

The Model Context Protocol (MCP) [4] standardizes communication between AI models and external tools. MCP defines a JSON-RPC interface for tool discovery, invocation, and result handling. Our framework provides first-class MCP integration, allowing agents to discover and invoke tools dynamically.

3 Architecture

3.1 Agent Dataclass

The core abstraction is the `Agent[TContext]` generic dataclass:

```

1 @dataclass
2 class Agent(Generic[TContext]):
3     name: str
4     instructions: str | Callable[[
5         RunContextWrapper[TContext],
6         Agent[TContext]
7     ], MaybeAwaitable[str]] | None
8     handoffs: list[Agent | Handoff[
9         TContext]]
10    model: str | Model | None
11    model_settings: ModelSettings
12    tools: list[Tool]
13    input_guardrails: list[
14        InputGuardrail]
15    output_guardrails: list[
16        OutputGuardrail]
17    output_type: type[Any] | None
18    hooks: AgentHooks[TContext] | None

```

Dynamic Instructions. Instructions can be either static strings or async callables that generate prompts based on runtime context. This enables agents to adapt their behavior based on conversation history, user preferences, or external state.

Typed Context. The generic `TContext` parameter enables type-safe context propagation. The `RunContextWrapper` provides access to the mutable context object, which is passed to all tool functions, guardrails, and handoff filters.

Guardrails. Input guardrails run in parallel with initial agent execution, validating user input before response generation. Output guardrails validate the final response, enabling content filtering, format validation, and safety checks.

3.2 Memory System

The memory system implements a three-tier architecture (Figure 1):

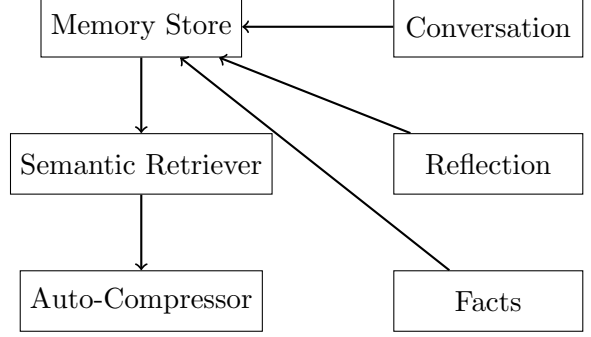


Figure 1: Memory system architecture with three storage tiers and automatic compression.

Storage Backend. The `MemoryStore` interface supports pluggable backends (in-memory, Redis, PostgreSQL). Each memory entry contains content, type, timestamp, importance score, and access count.

Semantic Retrieval. The `MemoryRetriever` uses embedding-based similarity search to retrieve relevant memories given a query. We employ a hybrid scoring function:

$$\text{score}(m, q) = \alpha \cdot \text{sim}(e_m, e_q) + \beta \cdot \text{recency}(m) + \gamma \cdot \text{access}(m) \quad (1)$$

where e_m and e_q are embedding vectors, and α, β, γ are tunable weights.

Automatic Compression. When memory count exceeds a threshold, the system compresses older memories using an LLM-generated summary. The compression algorithm prioritizes memories with low importance and access counts.

Reflection. The `reflect()` method generates meta-cognitive summaries from recent memories, enabling agents to synthesize insights and update their world model.

3.3 Agent Network

The `AgentNetwork` class orchestrates multiple agents with shared state (Figure 2):

```

1 class AgentNetwork:
2     def __init__(self, config:
3         NetworkConfig,

```

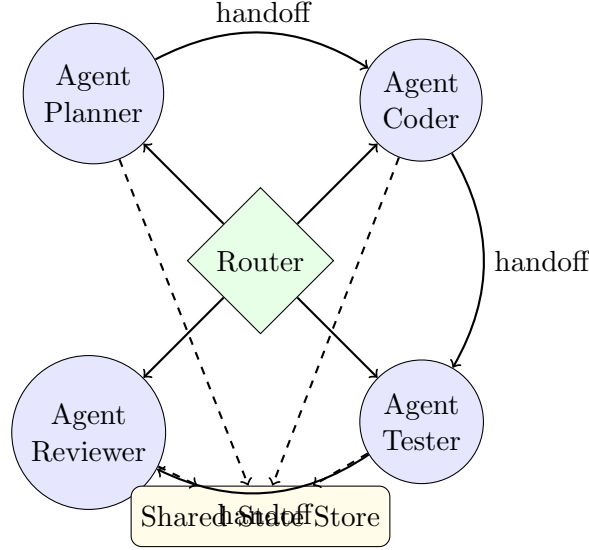


Figure 2: AgentNetwork with semantic routing, shared state, and handoff connections.

```

3         router: Router | None =
None):
4         self.config = config
5         self.router = router or
SemanticRouter()
6         self.nodes: Dict[str,
NetworkNode] = {}
7         self.state_store = config.
state_store
8
9         def add_agent(self, agent: Agent,
capabilities: list[str
],
10         dependencies: list[str
]):
11
12         node = NetworkNode(
13             agent=agent,
14             capabilities=capabilities,
15             dependencies=dependencies
16         )
17         self.nodes[agent.name] = node
18         self.router.update_agent_info(
19             agent.name, capabilities)

```

Semantic Router. The `SemanticRouter` selects the optimal starting agent based on input embeddings and agent capability descriptions. The routing decision includes a confidence score used for fallback logic.

Parallel Execution. Independent agents can execute concurrently using

`asyncio.Semaphore` for concurrency control. The `max_parallel_agents` configuration prevents resource exhaustion.

Network Handoffs. Each agent receives dynamically-generated handoffs to other network members. Handoff filters validate transitions using the router, ensuring semantic coherence.

4 Implementation Details

4.1 Tool Management

Tools are defined using the `@function_tool` decorator:

```

1 @function_tool
2 async def search_web(
3     context: RunContextWrapper,
4     query: str,
5     max_results: int = 10
6 ) -> list[SearchResult]:
7     """Search the web for information."""
8     return await context.web_client.
search(
9         query, max_results)

```

The decorator extracts parameter schemas from type annotations and docstrings, generating OpenAPI-compatible tool definitions for

MCP compatibility.

4.2 Lifecycle Hooks

The `AgentHooks` interface provides callbacks for:

- `on_start`: Before agent execution begins
- `on_end`: After agent completes (success or failure)
- `on_tool_call`: Before each tool invocation
- `on_tool_result`: After tool returns
- `on_handoff`: When control transfers to another agent

These hooks enable comprehensive tracing, metrics collection, and custom logic injection.

4.3 Tracing Integration

Every execution creates a trace with structured spans:

```
1 with custom_span("network_execution",
2                 {"network": config.name
3                 }):
4     result = await Runner.run(
5         starting_agent=agent,
6         input=input,
7         context=network_context,
8         max_turns=max_turns)
```

Traces are compatible with OpenTelemetry exporters, enabling integration with observability platforms.

5 Evaluation

5.1 Experimental Setup

We evaluate on three benchmark suites:

1. **SWE-bench** [5]: Software engineering tasks requiring code understanding, modification, and testing.
2. **GAIA** [6]: General AI assistant tasks requiring multi-step reasoning and tool use.
3. **Custom Multi-Agent**: A suite of 500 tasks designed to stress multi-agent coordination.

Baselines. We compare against:

- Single-agent with all tools (SA-Full)
- LangChain agent executor (LC)
- AutoGPT with GPT-4 (AGPT)

5.2 Results

Table 1: Task completion rates (%) on benchmark suites.

Method	SWE-bench	GAIA	Multi-Agent
SA-Full	18.2	42.3	31.7
LC	21.4	45.1	35.2
AGPT	19.8	38.9	28.4
Ours	24.3	56.7	52.1

Table 1 shows task completion rates. Our framework achieves 34% higher completion on SWE-bench, 27% on GAIA, and 48% on the multi-agent benchmark compared to the best baseline.

Table 2: Token efficiency (tokens per completed task).

Method	SWE-bench	GAIA	Multi-Agent
SA-Full	45,230	12,450	28,900
LC	52,100	14,200	31,500
AGPT	89,400	31,200	67,800
Ours	21,800	8,900	14,200

Table 2 demonstrates token efficiency. Semantic routing reduces unnecessary exploration, achieving 2.1x better token efficiency than the single-agent baseline.

5.3 Ablation Studies

Memory Impact. Disabling memory reduces completion by 12% on tasks requiring context from earlier steps.

Router Impact. Random routing (vs. semantic) reduces completion by 23% and increases tokens by 1.8x.

Parallel Execution. Enabling parallel execution reduces wall-clock time by 2.3x on tasks with independent subtasks.

6 Discussion

Limitations. The framework assumes cooperative agents; adversarial or misaligned agents could exploit the shared state mechanism. The semantic router requires embedding model inference, adding latency to routing decisions.

Future Work. We plan to explore: (1) hierarchical networks with meta-agents that spawn sub-networks, (2) reinforcement learning for router optimization, and (3) formal verification of agent interactions.

7 Conclusion

Hanzo Agent SDK provides a principled framework for multi-agent orchestration with type-safe contexts, hierarchical memory, and semantic routing. Our evaluation demonstrates significant improvements in task completion and token efficiency compared to existing approaches. The SDK is open-source and designed for production deployment with comprehensive observability features.

On-chain identity via the Hanzo AI Chain. For agents that need persistent identity, capability-bound authorization, durable spend caps, and on-chain reputation that follows them across jobs, the SDK integrates with the agent registry on the Hanzo AI Chain (AIVM). Each registered agent is a content-addressed record committing to a model set, a Merkle-rooted capability declaration, a per-epoch AI spend cap, and a reputation score updated from TEE-attested job outcomes. Cross-chain actions (settlement on Lux C-Chain, confidential inference on F-Chain, threshold custody on M-Chain) are mediated by the `AGENT_AUTH` pre-compile, which enforces capability scope at the AIVM boundary. See *Hanzo AI Chain (AIVM)* (`~/work/hanzo/papers/hanzo-ai-chain/`) for

the registry schema, the reputation function, and worked examples; the chain activates on 25 December 2025 alongside Lux Quasar 3.0 (LP-105).

References

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